WEEK – 11

1# Create an inter University Database with the following relations. Include at least four attributes for each table.

- University
- Department
- Program
- Course
- Syllabus
- Faculty (Teacher)

a) Create ER Diagram and Schema diagram for the above system.

*Attached at end

b) Create above tables and mention primary keys and foreign keys. Also create secondary index for each table

```sql
CREATE TABLE UNIVERSITY (
    Univ_ID INT PRIMARY KEY,
    Univ_Name VARCHAR(100),
    Location VARCHAR(50),
    Established_Year INT
);
CREATE INDEX idx_univ_name ON UNIVERSITY (Univ_Name);
```

Results | Explain | Describe | Saved SQL | History

Index created.

0.06 seconds

```sql
CREATE TABLE DEPARTMENT (
    Dept_ID INT PRIMARY KEY,
    Dept_Name VARCHAR(100),
    Univ_ID INT,
    HOD_Name VARCHAR(100),
    FOREIGN KEY (Univ_ID) REFERENCES UNIVERSITY(Univ_ID)
);
CREATE INDEX idx_dept_name ON DEPARTMENT(Dept_Name);

CREATE TABLE PROGRAM (
    Program_ID INT PRIMARY KEY,
    Program_Name VARCHAR(100),
    Dept_ID INT,
    Duration INT,
    FOREIGN KEY (Dept_ID) REFERENCES DEPARTMENT(Dept_ID)
);
CREATE INDEX idx_program_name ON PROGRAM (Program_Name);
```

```sql
CREATE TABLE COURSE (
    Course_ID INT PRIMARY KEY,
    Course_Name VARCHAR(100),
    Program_ID INT,
    Credits INT,
    FOREIGN KEY (Program_ID) REFERENCES PROGRAM(Program_ID)
);
CREATE INDEX idx_course_name ON COURSE(Course_Name);

CREATE TABLE SYLLABUS (
    Syllabus_ID INT PRIMARY KEY,
    Course_ID INT,
    Topic_Name VARCHAR(100),
    Description VARCHAR(200),
    FOREIGN KEY (Course_ID) REFERENCES COURSE(Course_ID)
);
CREATE INDEX idx_topic ON SYLLABUS(Topic_Name);
```

```sql
CREATE TABLE FACULTY (
    Faculty_ID INT PRIMARY KEY,
    Faculty_Name VARCHAR(100),
    Specialization VARCHAR(100),
    Dept_ID INT,
    FOREIGN KEY (Dept_ID) REFERENCES DEPARTMENT(Dept_ID)
);
CREATE INDEX idx_specialization ON FACULTY(Specialization);
```

b) Insert at least 5 relevant records in each of the created tables.

```sql
INSERT INTO UNIVERSITY VALUES
(1, 'AMU', 'Aligarh', 1920),
(2, 'JNU', 'Delhi', 1969),
(3, 'University of Jammu', 'Jammu', 1965),
(4, 'Delhi University', 'Delhi', 1922),
(5, 'Mumbai University', 'Mumbai', 1857);

INSERT INTO DEPARTMENT VALUES
(101, 'Computer Science', 1, 'Dr A'),
(102, 'IT', 2, 'Dr B'),
(103, 'Electronics', 3, 'Dr C'),
(104, 'Computer Science', 4, 'Dr D'),
(105, 'Management', 5, 'Dr E');

INSERT INTO PROGRAM VALUES
(201, 'MCA', 101, 3),
(202, 'BTech', 102, 4),
(203, 'MSc CS', 103, 2),
(204, 'MCA', 104, 3),
(205, 'MBA', 105, 2);
```

```sql
INSERT INTO COURSE VALUES
(301, 'Computer Architecture', 201, 4),
(302, 'DBMS', 201, 4),
(303, 'Network Security', 202, 3),
(304, 'Operating Systems', 204, 4),
(305, 'AI', 203, 3);

INSERT INTO SYLLABUS VALUES
(401, 301, 'CPU Design', 'Basics of CPU architecture'),
(402, 302, 'SQL Basics', 'Introduction to SQL'),
(403, 303, 'Encryption', 'Information Security concepts'),
(404, 304, 'Process Management', 'OS concepts'),
(405, 305, 'Machine Learning', 'AI fundamentals');

INSERT INTO FACULTY VALUES
(501, 'Dr X', 'Information Security', 101),
(502, 'Dr Y', 'Networks', 102),
(503, 'Dr Z', 'AI', 103),
(504, 'Dr W', 'Information Security', 104),
(505, 'Dr V', 'DBMS', 105);
```

d) Write following SQL queries based on above created database:

i) List of Universities situated at Mumbai.

```sql
SELECT * FROM UNIVERSITY
WHERE Location = 'Mumbai';
```

Results | Explain | Describe | Saved SQL | History

UNIV_ID	UNIV_NAME	LOCATION	ESTABLISHED_YEAR
5	Mumbai University	Mumbai	1857

1 rows returned in 0.02 seconds Download

ii) List of all the programs run by the AMU.

```sql
SELECT p.*
FROM PROGRAM p
JOIN DEPARTMENT d ON p.Dept_ID = d.Dept_ID
JOIN UNIVERSITY u ON d.Univ_ID = u.Univ_ID
WHERE u.Univ_Name = 'AMU';
```

PROGRAM_ID	PROGRAM_NAME	DEPT_ID	DURATION
201	MCA	101	3

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iii) Find out all the departments of JNU.

```sql
SELECT d.*
FROM DEPARTMENT d
JOIN UNIVERSITY u ON d.Univ_ID = u.Univ_ID
WHERE u.Univ_Name = 'JNU';
```

Results | Explain | Describe | Saved SQL | History

DEPT_ID	DEPT_NAME	UNIV_ID	HOD_NAME
102	IT	2	Dr B

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iv) List of all Programs run by University of Jammu

```sql
SELECT p.*
FROM PROGRAM p
JOIN DEPARTMENT d ON p.Dept_ID = d.Dept_ID
JOIN UNIVERSITY u ON d.Univ_ID = u.Univ_ID
WHERE u.Univ_Name = 'University of Jammu';
```

Results | Explain | Describe | Saved SQL | History

PROGRAM_ID	PROGRAM_NAME	DEPT_ID	DURATION
203	MSc CS	103	2

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v) List of Universities that run Program "MCA".

```sql
SELECT DISTINCT u.Univ_Name
FROM UNIVERSITY u
JOIN DEPARTMENT d ON u.Univ_ID = d.Univ_ID
JOIN PROGRAM p ON d.Dept_ID = p.Dept_ID
WHERE p.Program_Name = 'MCA';
```

Results | Explain | Describe | Saved SQL | History

UNIV_NAME
AMU
Delhi University

2 rows returned in 0.04 seconds Download

vi) List of Courses of "MCA" run by AMU.

```sql
SELECT c.*
FROM COURSE c
JOIN PROGRAM p ON c.Program_ID = p.Program_ID
JOIN DEPARTMENT d ON p.Dept_ID = d.Dept_ID
JOIN UNIVERSITY u ON d.Univ_ID = u.Univ_ID
WHERE p.Program_Name = 'MCA'
AND u.Univ_Name = 'AMU';
```

Results | Explain | Describe | Saved SQL | History

COURSE_ID	COURSE_NAME	PROGRAM_ID	CREDITS
301	Computer Architecture	201	4
302	DBMS	201	4

2 rows returned in 0.03 seconds Download

vii) List of Faculties specialized in "Information Security" across different universities.

```sql
SELECT f.Faculty_Name, u.Univ_Name
FROM FACULTY f
JOIN DEPARTMENT d ON f.Dept_ID = d.Dept_ID
JOIN UNIVERSITY u ON d.Univ_ID = u.Univ_ID
WHERE f.Specialization = 'Information Security';
```

Results | Explain | Describe | Saved SQL | History

FACULTY_NAME	UNIV_NAME
Dr X	AMU
Dr W	Delhi University

2 rows returned in 0.04 seconds Download

viii) Syllabus of "Computer Architecture" of different Universities.

```sql
SELECT u.Univ_Name, c.Course_Name, s.Topic_Name, s.Description FROM UNIVERSITY u
JOIN DEPARTMENT d ON u.Univ_ID = d.Univ_ID
JOIN PROGRAM p ON d.Dept_ID = p.Dept_ID
JOIN COURSE c ON p.Program_ID = c.Program_ID
JOIN SYLLABUS s ON c.Course_ID = s.Course_ID
WHERE c.Course_Name = 'Computer Architecture';
```

Results | Explain | Describe | Saved SQL | History

UNIV_NAME	COURSE_NAME	TOPIC_NAME	DESCRIPTION
AMU	Computer Architecture	CPU Design	Basics of CPU architecture

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ix) List of all faculties with their specializations of Department of Computer Science of Delhi University.

```sql
SELECT f.Faculty_Name, f.Specialization
FROM FACULTY f
JOIN DEPARTMENT d ON f.Dept_ID = d.Dept_ID
JOIN UNIVERSITY u ON d.Univ_ID = u.Univ_ID
WHERE d.Dept_Name = 'Computer Science'
AND u.Univ_Name = 'Delhi University';
```

Results | Explain | Describe | Saved SQL | History

FACULTY_NAME	SPECIALIZATION
Dr W	Information Security

1 rows returned in 0.03 seconds Download

x) Find the University running maximum programs

```sql
SELECT Univ_Name, Total_Programs
FROM (
    SELECT u.Univ_Name, COUNT(p.Program_ID) AS Total_Programs
    FROM UNIVERSITY u
    JOIN DEPARTMENT d ON u.Univ_ID = d.Univ_ID
    JOIN PROGRAM p ON d.Dept_ID = p.Dept_ID
    GROUP BY u.Univ_Name
    ORDER BY COUNT(p.Program_ID) DESC
)
WHERE ROWNUM = 1;
```

Results | Explain | Describe | Saved SQL | History

UNIV_NAME	TOTAL_PROGRAMS
AMU	1

1 rows returned in 0.03 seconds Download